



Research article

Genome-wide identification, characterization and expression profiling of *LIM* family genes in *Solanum lycopersicum* L.

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ABSTRACT

LIM domain proteins, some of which have been shown to be actin binding proteins, are involved in various developmental activities and cellular processes in plants. To date, the molecular defense-related functions of *LIM* family genes have not been investigated in any solanaceous vegetable crop species. In this study, we identified 15 *LIM* family genes in tomato (*Solanum lycopersicum* L.) through genome-wide analysis and performed expression profiling in different organs of tomato, including fruits at six different developmental stages. We also performed expression profiling of selected tomato *LIM* genes in plants under ABA, drought, cold, NaCl and heat stress treatment. The encoded proteins of the 15 tomato *LIM* genes were classified into two main groups, i.e., proteins similar to cysteine-rich proteins and plant-specific DAR proteins, based on differences in functional domains and variability in their C-terminal regions. The DAR proteins contain a so far poorly characterized zinc-finger-like motif that we propose to call DAR-ZF. Six of the 15 *LIM* genes were expressed only in flowers, indicating that they play flower-specific roles in plants. The other nine genes were expressed in all organs and at various stages of fruit development. *SlβLIM1b* was expressed relatively highly at the later stage of fruit development, but three other genes, *SlWLIM2a*, *SIDAR2* and *SIDAR4*, were expressed at the early stage of fruit development. Seven genes were induced by ABA, five by cold, seven by drought, eight by NaCl and seven by heat treatment respectively, indicating their possible roles in abiotic stress tolerance. Our results will be useful for functional analysis of *LIM* genes during fruit development in tomato plants under different abiotic stresses.

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1. Introduction

LIM proteins trigger actin bundle formation during plant growth and development (Thomas et al., 2006, 2008; Wang et al., 2008;

Papuga et al., 2010; Han et al., 2013). These actin bundles are key components of the cytoskeleton. *LIM* proteins are also involved in transcriptional modulation, as key genes of the phenylpropanoid pathway are regulated by NtLIM1 (Kawaoka et al., 2000). *WLIM1* in tobacco and six Arabidopsis *LIM* genes are crucial for the formation of actin filaments, which protect the actin cytoskeleton from disassembly (Thomas et al., 2006; Papuga et al., 2010). *LIM* genes in cotton play dual roles in cell elongation and secondary wall formation (Han et al., 2013). As the *LIM* gene family is involved in transcriptional regulation and other developmental activities in various plant systems, characterizing this gene family is important for elucidating the crucial roles of *LIM* proteins in plant

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development and defense (Park et al., 2014).

The acronym LIM was derived from the names of three animal homeodomain proteins, LIN11, ISL1 and MEC3, which each contain two LIM domains (Freyd et al., 1990; Karlsson et al., 1990; Way and Chalfie, 1988). LIM domain proteins have a double-zinc finger motif of the sequence [C-X₂-C-X_{16–23}-H-X₂-C]-X₂-[C-X₂-C-X_{16–23}-C-X₂-C(D,H)], which contains cysteine-histidine residues in the conserved region that function as a protein-interacting domain (Schmeichel and Beckerle, 1994). Plant species contain two distinct LIM domain protein families. The first type contains two LIM domains similar to those found in the cysteine-rich protein (CRP) family, and the second type contains one LIM domain with or without a ubiquitin interacting motif (UIM) domain. In the first type of LIM domain protein, two LIM domains are separated by an inter-LIM spacer and the C-terminus is relatively short and variable (Eliasson et al., 2000; Arnaud et al., 2007). The second type of LIM domain protein is characterized by the presence of a single LIM domain, with or without one or two UIMs, one DUF3633 domain (a domain of unknown function), a variable N-terminus and a conserved C-terminus (Li et al., 2008; Xia et al., 2013; Zhao et al., 2014). *Arabidopsis thaliana* and rice contain six CRP-like LIM genes, whereas poplar contains 12 (Arnaud et al., 2007). In *Arabidopsis*, the CRP-like LIM proteins influence actin cytoskeleton organization and are involved in binding to actin filaments (Papuga et al., 2010). LIM domain proteins are also involved in various cellular activities including gene transcription regulation, cell lineage specification and organ development (Velyvis and Qin, 2005). The regulatory role of these actin-binding proteins (ABPs) is accomplished by direct interaction with monomeric and/or polymerized actin to promote various actin filament activities (Winder and Ayscough, 2005).

Plants have evolved many mechanisms for surviving under unsuitable environmental conditions. In most cases, the plant cytoskeleton acts as a key regulator of various stress responses. The cytoskeleton plays a dynamic role in almost every intracellular activity including cell division, cell movement, cell morphogenesis and signal transduction, as well as plant defense responses (Wasteneys and Yang, 2004; Wang et al., 2011). It is therefore important to investigate the functional behavior of LIM family genes, which are involved in cytoskeleton reorganization and signal transduction pathways. Experimental evidence suggests that CRP-like proteins and some other structurally related plant proteins function like ABPs (Winder and Ayscough, 2005; Grubinger and Gimona, 2004; Tran et al., 2005; Thomas et al., 2006). DA1 and

DA1-related (DAR), which are also encoded by LIM family genes, play various roles in plants, such as controlling the size of seeds, organs and root meristems, as well as in disease resistance signaling and ABA/cold responses (Li et al., 2008; Xia et al., 2013; Yang et al., 2010). Functional analysis of proteins produced by LIM family genes suggests that they are involved in growth, development and abiotic stress tolerance in plants. However, their roles have not been functionally validated, except in some recent investigations in *Brassica rapa* (Park et al., 2014) and *Cicer arietinum* (Srivastava and Verma, 2015).

Since the LIM gene family plays regulatory roles in developmental processes and defense response in plants, we wanted to identify LIM genes in tomato (*Solanum lycopersicum*) and to investigate their involvement in different biological processes, including organ and fruit development and stress tolerance. Tomato is one of the most common vegetables worldwide and is a good model system for studying climacteric fruit development and ripening. The recently published tomato genome sequence has opened the door to genome-wide analysis of specific gene families (Tomato Genome Consortium, 2012). In the present study, we identified 15 tomato LIM genes using the Sol Genomics database and performed phylogenetic analysis to examine their homology to other plant LIM genes. We also examined the tissue-specific expression patterns of these genes in different organs, as well as during fruit development. Finally, we examined the expression profiles of these genes under several abiotic stresses to investigate their roles in stress tolerance in tomato.

2. Materials and methods

2.1. Sequence analysis of tomato LIM genes and the corresponding LIM proteins

Fifteen LIM genes were identified by BLAST searches in the Sol Genomics database (<http://solgenomics.net/>) using 'Interpro LIM Domain Annotation IPR001781' as a reference sequence (Table 1). The LIM protein sequences from *Arabidopsis thaliana*, *Brassica rapa* and *Oryza sativa* were obtained from TAIR (<http://arabidopsis.org>), BRAD (<http://brassicadb.org>) and the TIGR plant genomic group (<http://plantgenomics.tigr.org/>), respectively. *Populus trichocarpa*, *Helianthus annuus* and *Nicotiana tabacum* protein sequences were obtained from the NCBI database. The physicochemical properties of the tomato LIM proteins, including sequence length, molecular weight and isoelectric point, were identified using the Prototaram

Table 1

In silico analysis of LIM genes and corresponding proteins identified from the *Solanum lycopersicum* database.^a

Sl. no.	Locus name	ORF (bp)	Chromosome location (no.)	Protein				
				Length (aa)	LIM domain start-end (aa)	Mol. wt. (kDa)	pI	GRAVY
1	Solyc01g094320	672	01	224	9–61 and 105–157	24.63	5.98	−0.685
2	Solyc02g089930	1548	02	516	146–198	57.75	5.48	−0.608
3	Solyc03g079910	1650	03	550	186–238	62.42	7.03	−0.461
4	Solyc03g114000	570	03	190	10–62 and 108–160	21.27	9.29	−0.637
5	Solyc04g077780	582	04	194	9–61 and 109–161	21.62	8.94	−0.465
6	Solyc04g079840	1461	04	487	124–176	55.41	5.86	−0.591
7	Solyc05g049870	687	05	229	9–61 and 105–157	25.16	5.79	−0.733
8	Solyc05g052780	570	05	190	9–61 and 107–159	20.82	9.14	−0.497
9	Solyc05g052910	1515	05	505	140–192	56.96	8.28	−0.584
10	Solyc06g071310	540	06	180	10–62 and 107–159	20.23	8.97	−0.669
11	Solyc08g007940	621	08	207	9–61 and 102–154	22.52	7.50	−0.616
12	Solyc08g080740	585	08	195	9–61 and 103–155	21.44	9.10	−0.464
13	Solyc10g017520	639	10	213	9–61 and 105–157	23.60	5.92	−0.533
14	Solyc11g044740	582	11	194	9–61 and 109–161	21.68	8.63	−0.478
15	Solyc12g088320	564	12	188	9–61 and 106–158	21.07	9.11	−0.613

Abbreviations: ORF, open reading frame; bp, base pair; aa, amino acid; kDa, kilo Dalton; pI, isoelectric point; MW, molecular weight; GRAVY, Grand average of hydropathicity.

^a Sol Genomics database (SGN, <http://solgenomics.net>).

(<http://expasy.org/tools/protparam.html>) tool. Gene Structure Display Server 2.0 was used to analyze the exon-intron distributions (<http://gsds2.cbi.pku.edu.cn/>). Open reading frames (ORFs) were analyzed using the ORF finder (<http://www.ncbi.nlm.nih.gov/gorf/gorf.html>) tool. The SMART web tool was used to identify the LIM domains (http://smart.embl.de/smart/set_mode.cgi?GENOMIC=1), and the Genedoc protein alignment tool (www.nrbsc.org/gfx/genedoc/ebinet.htm) was used to analyze multiple protein sequences. The MEME suite motif search tool was used to identify the motifs in protein sequences. Homology analyses of tomato LIM proteins with other plant LIM proteins were conducted using the NCBI protein BLAST tool. Sequence homology among the 15 LIM proteins was tested using CLUSTALW (<http://www.genome.jp/tools/clustalw/>). Two separate phylogenetic trees were constructed using MEGA6.0 software (<http://www.megasoftware.net>) according to the neighbor-joining method based on the evolutionary relatedness of tomato LIM proteins using both full-length deduced polypeptide sequences and conserved LIM domain sequences only of LIM genes (Supplementary data, Saitou and Nei, 1987; Tamura et al., 2013).

2.2. Chromosomal locations and gene duplication analysis of the tomato LIM genes

The genomic distribution of tomato LIM genes was determined by performing BLASTN analysis of the cDNA sequence of each gene against the Sol Genomics database (<http://www.cornel.edu/tools/blast>). The position of each gene on the chromosomes was drawn based on the MAP chart results (<http://www.wageningenur.nl/en/show/Mapchart.html>). All tomato LIM genes were used as queries with the BLAST tool from NCBI (<http://www.ncbi.nlm.nih.gov/blast>) to determine the identity and percentage similarity among sequences.

2.3. Promoter analysis

The promoter sequences of each gene were analyzed using the PlantCARE database (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>). Nucleotide sequences of 1500 base pairs prior to the coding DNA sequence of each target gene were selected to search for promoter sequences.

2.4. Plant material collection and preparation

S. lycopersicum cv. Ailsa Craig plants were grown in soil in a growth room at the Department of Horticulture of Sunchon National University, South Korea at 25 °C under a 16 h light/8 h dark cycle. Fresh roots, stems and leaves were harvested from 28-day-old plants to study the tissue-specific expression patterns of LIM genes. The plants were then transferred to a greenhouse for further growth. Flowers were collected immediately after blooming, and fruit samples were collected at six different developmental stages: i) young fruits approximately one week after the date of pollination and approximately 0.8–1.0 cm in diameter, ii) immature fruits approximately 20 days after the date of pollination and approximately 2 cm in diameter, iii) mature green fruits 45 days after the date of pollination, iv) fruits at the breaker stage, when the colour of green mature fruits changed to faint yellow-orange, v) fruits 5 days after the breaker stage and vi) fruits 10 days after the breaker stage. The samples were frozen in liquid nitrogen immediately after collection and stored at –80 °C for RNA isolation.

To impose abiotic stress treatments, 28-day-old seedlings of uniform growth and development were selected. The abiotic stress treatments were as follows: ABA, drought, cold, NaCl and heat. For each treatment, three individual plants were selected as three

independent biological replicates. For ABA treatment, each tomato seedling was sprayed with 100 µM ABA. For drought treatment, whole tomato seedlings were gently pulled from the soil, washed carefully with water to remove the adhering soil and incubated on a dry paper towel at 25 ± 1 °C; no additional water was supplied during the treatment period. For cold and heat treatment, potted tomato seedlings were incubated at 4 °C and 40 °C, respectively. Salinity treatment was applied by submerging the roots of the tomato seedlings in a 200 mM NaCl solution. Leaf samples were collected at 0, 1, 3, 6, 12 and 24 h time intervals for all treatments. For all abiotic stress treatments, samples collected at 0 h were used as the control. All samples were immediately frozen in liquid nitrogen and stored at –80 °C for RNA isolation.

2.5. RNA extraction

RNA was extracted from fresh roots, leaves, stems, flowers and fruits at different developmental stages and from stress-treated samples using an RNeasy mini kit (Qiagen, Valencia, USA). Genomic DNA contaminants were removed by treating the RNA with RNase-free DNase (Promega, USA). A NanoDrop® 1000 Spectrophotometer (Wilmington, USA) was used to measure the RNA. A Superscript® III First-Strand Synthesis kit (Invitrogen, USA) was used to synthesize single-stranded cDNA from RNA following the manufacturer's protocol.

2.6. RT-PCR analysis

RT-PCR of tomato samples was performed using an AMV one-step RT-PCR kit (Takara, Shiga, Japan). Gene-specific primers were used for the RT-PCR analysis (Supplementary Table 1). Primer sequences were designed using Primer3 software (<http://frodo.wi.mit.edu/primer3/input.htm>), and cDNA from different organs was used as DNA template. The PCR mixture (20 µl) contained 1 µl each of forward and reverse primers, 8 µl Emerald PCR Master Mix (Takara, Shiga, Japan) and 9 µl double-distilled H₂O. The PCR conditions were as follows: pre-denaturation for 5 min at 94 °C, amplification for 30 cycles at 94 °C for 30 s, annealing for 30 s at 58 °C and extension for 1 min at 72 °C, with a final extension at 72 °C for 5 min. *EF1a* primers from *S. lycopersicum* (F: TCAGG-TAAGGAAGCTTGAGAAGGAGCCT, R: AGTTCACCTCCCTTCTCTGGG-CAG) were used as a control for both RT-PCR and qPCR analysis (Aoki et al., 2010).

2.7. qPCR expression analysis

The qPCR analysis was performed with 1 µl cDNA in a 20 µl mixture using an iTaq™ SYBR®Green PCR kit (Qiagen, California, USA). Newly designed specific primers were used to amplify 15 LIM genes for qPCR analysis (Supplementary Table 2). The relative expression levels were calculated compared to genes expressed in all organs, including leaf samples. The qPCR conditions were as follows: 95 °C for 15 min followed by 40 cycles at 94 °C for 20 s, 58 °C for 20 s and 72 °C for 40 s. An individual replicate against each treatment was read at the end of each cycle. The LightCycler96 (Roche, Germany) was used to amplify and record the C_q value of each sample. The widely used 2^{–ΔΔC_q} method was used to estimate the relative expression level of each gene against each treatment (Schmittgen and Livak, 2008). Analysis of variance of the relative expression level of each gene at different sampling points under each abiotic stress treatment was carried out following a generalized linear model using MINITAB statistical software (Minitab Inc., State College, PA, USA). Significant differences in mean values at different sampling times were determined by Tukey's pairwise comparison tests, as indicated by different letters in the figures.

3. Results

3.1. Identification of tomato LIM proteins and LIM genes

The predicted lengths of the 15 tomato LIM proteins identified ranged from 180 to 550 amino acids, and the molecular weights ranged from 20.23 to 62.42 KDa (Table 1). The isoelectric points varied from 5.48 to 9.29, indicating that these proteins include both those acidic and those basic in nature (Table 1). The grand average of hydropathicity (GRAVY) ranged from -0.461 to -0.733 , indicating that all deduced polypeptides are hydrophilic in nature (Table 1). Analysis of the domain structure revealed that 11 of the 15 proteins contained two LIM domains in the conserved region located primarily between positions 9–61 and 102–160, whereas the four remaining proteins contained one LIM domain with or without a UIM domain and DUF3633 domain (Table 1). DUF3633 domain family proteins that have LIM domains are approximately 210 amino acids in length (www.ncbi.nlm.nih.gov/).

3.2. Phylogenetic classification of tomato LIM proteins

The 15 LIM proteins were initially divided into two subfamilies, based on the presence of CRP-like and DAR proteins (Fig. 1). The CRP-like proteins in plants were classified into two major groups: i) those expressed only in pollen (PLIM) and ii) those widely expressed in different organs (WLIM) (Fig. 1, Baltz et al., 1999; Eliasson et al., 2000; Papuga et al., 2010). β LIM is a newly classified group and the members of this group have wide expression patterns in *B. rapa* (Park et al., 2014). Based on phylogenetic analysis

(constructed using full-length peptide sequences), tomato LIM proteins 2, 3, 6 and 9 were classified into the DAR subfamily and the 11 other proteins were classified into the CRP-like subfamily (Fig. 1). The CRP-like subfamily included the PLIM1, PLIM2, β LIM1, WLIM1 and WLIM2 sub-groups based on their organ-specific expression patterns and similarity with other plant LIM proteins (Fig. 1, Supplementary Table 3). The 15 tomato LIM proteins were named based on their phylogenetic association (Fig. 1, Supplementary Table 3). Five tomato LIM proteins belong to the PLIM2 clade, and were called SIPLIM2a, SIPLIM2b, SIPLIM2c, since they cluster with tobacco PLIM2, and SI δ LIM2a and SI δ LIM2b, since they cluster with tobacco δ LIM2, respectively. Two tomato LIM proteins cluster in the WLIM1 subgroup and were designated as SIWLIM1a and SIWLIM1b, respectively. One LIM protein, SIWLIM2, fell within the WLIM2 group. Two tomato LIM proteins were clustered in the β LIM1 sub-group with other plant proteins and were designated as SI β LIM1a and SI β LIM1b, respectively. One protein clustered in the PLIM1 subgroup with sunflower and tobacco PLIM1 proteins and was designated as SIPLIM1. Four LIM proteins belong to the DA1/DAR clade along with the eight DA1 and DAR sequences from Arabidopsis, eight DAR sequences from Chinese cabbage and four DAR sequences from rice (Fig. 1). As these four proteins shared a close relationship with other plant DA1 and DAR proteins, they were designated as SIDAR1, 2, 3 and 4, respectively (Fig. 1, Supplementary Table 3). Of the seven moss LIM protein sequences, four belong to the DA1/DAR clade, and three to the CRP clade. A phylogenetic tree constructed using the conserved LIM domain only (Supplementary Fig. 4) resulted in a similar tree as that produced using the full-length protein sequences.

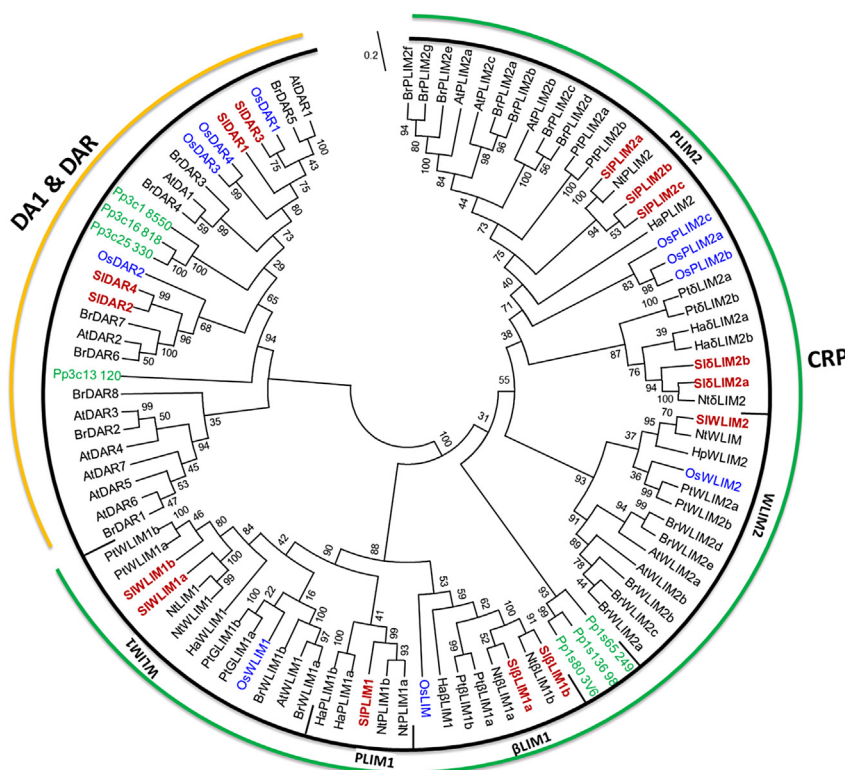


Fig. 1. Phylogenetic tree showing the relatedness of tomato LIM proteins to those of Arabidopsis, Chinese cabbage, rice, poplar, sunflower, tobacco and moss. Full-length polypeptide sequences were used to make the phylogenetic tree. The CRP-like and DAR LIM family was marked by green and orange arc line. Monocot rice and land plant moss LIM proteins were indicated by blue and green colour respectively. All dicot plant LIM proteins were indicated by black colour. Among the dicot plant LIM proteins were indicated by red colors. The numbers at the branches represented the percent bootstraps values based on 1000 replications. The scale represents the frequency of amino acid substitutions between sequences, as determined by the Poisson's evolutionary distance method following pair-wise deletion of amino acids. A species acronym was added before each LIM protein name: SI, *Solanum lycopersicum*; At, *Arabidopsis thaliana*; Pt, *Populus trichocarpa*; Br, *Brassica rapa*; Os, *Oryza sativa*; Ha, *Helianthus annuus*; Nt, *Nicotiana tabacum* and Pp, *Physcomitrella patens*. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

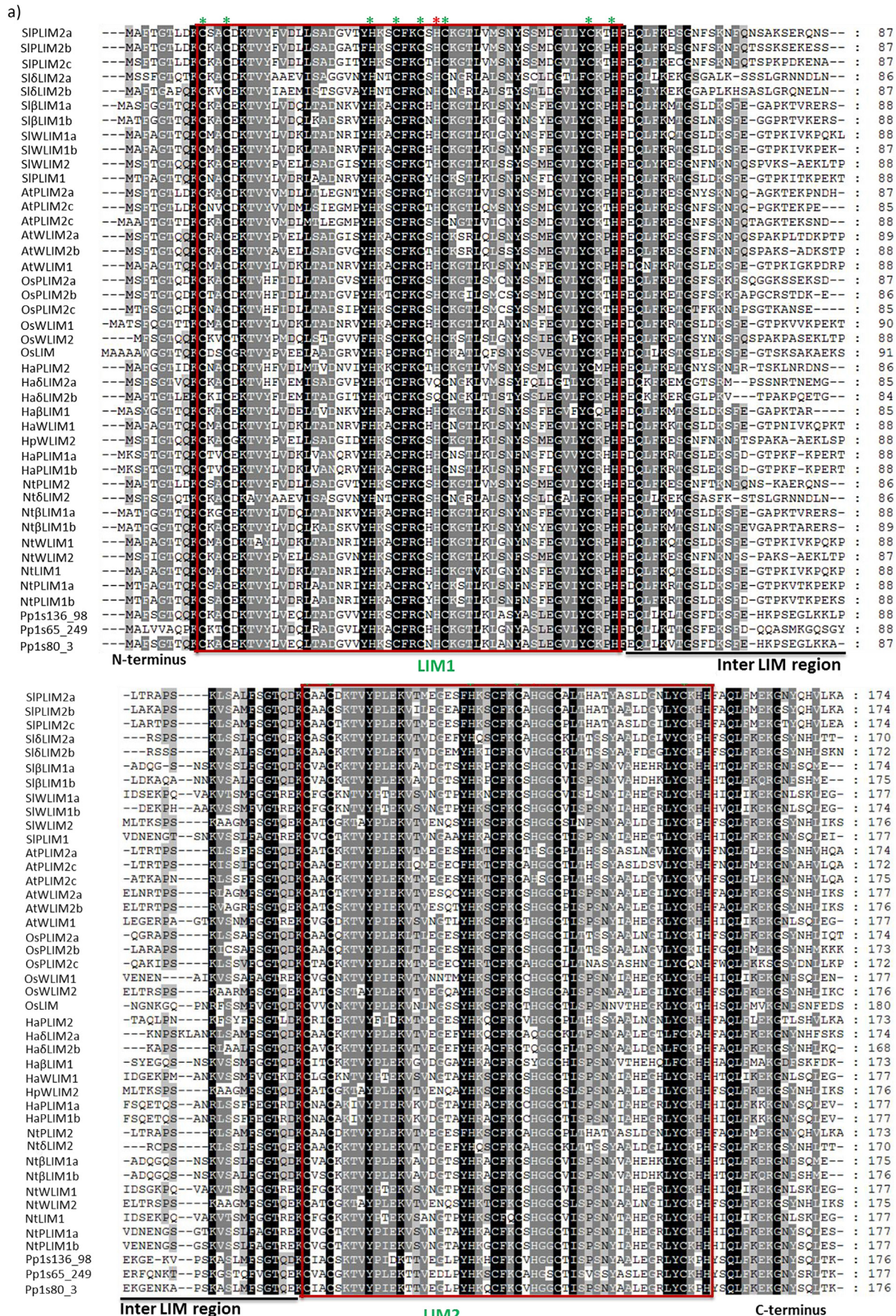


Fig. 2. Multiple sequence alignment of proteins of the a) CRP-like LIM family and b) plant-specific DAR family. Red boxes indicate the LIM domains, and blue boxes indicate the UIM domains. The DUF3633 domains are underlined. Conserved cysteine-histidine residues in the LIM domains were indicated by green asterisks. Red asterisks within the LIM domains indicate additional, not perfectly conserved histidine residues. (Note: AtDAR3 lacks the first three cysteine residue of the LIM domain.) The blue asterisks indicate a conserved block of about 80 residues, outlined in yellow, containing ten conserved cysteine-histidine residues. We refer to this new putative zinc finger domain as "DAR-ZF". Numbers in the right sight indicate amino acid positions. Identical amino acids are indicated by a dark black background and amino acids with >50% similarity is indicated by a light black background. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

S1DAR1	FKKEEIDRAIAIAIAEE	-----DQKSKVIV	ESEPHI	LDG	-----AIARAI	CESLNIV	ESSPPF	-----PRVYDGSPPFLSYISYFVYGR	GAGNAEIG	: 86			
S1DAR2	KEKEEEDHATAIAIAE	-----KRP-GYKCR	-----TINQ	-----DIARS	DHDLSSSY	PIXP	-----PPYAPISYAPWYFNPSYR	ONGHGLIG	: 78				
S1DAR3	-----IEEDIRAI	IAIAEE	-----QRQKIVID	SESC	LDG	-----DIARAI	CESLNIV	SPQHSVRNDHGGGVNGVNGYFPHVYFYSASF	CAGSTTEIG	: 91			
S1DAR4	S1DAR1	FKKEEIDRAIAIAE	LD	-----RRRQGYK	-----TINQ	-----DIARS	CDSN	SSSHI	-----PKYAPISYAPWYFNPSYR	CAGSKNLI	: 76		
AtCDAR1	FKKEEIECAIAIAE	SEEHVIP	DQKRII	EYK	SLDG	-----DIARAI	CESLNIV	SS	-----PRVYDGNLQIPYFLIPSSHR	YGGVAGT	: 86		
AtCDAR2	FKKEEIDRAIAIAE	-----KRPQYS	-----MANKR	-----DIARS	PPHGGN	SSPF	-----FPYEPYSQ	-----YR-RKQPS	CAGSKSLIG	: 72			
AtCDAR3	KIEIRRVKES	PKARAEKRR	LESEKQKRIQVDD	-----	-----	-----	-----	-----	-----	: 76			
AtCDAR4	ELISLIESK	RIKALG	EDIKRADE	CKRSKDHSE	-----IESNEKE	-----DQDRADE	EDIKRS	-----KDVVEE	-----VNRQPS	TDGKSEIG	: 76		
AtCDAR5	FISVLEENW	IGHL	NGGNGEN	DAVIL	CKL	-----	-----	-----	-----	: 81			
AtCDAR6	-----VKKKCE	PL	LNLMV	QPEENN	SSIS	-----	-----	-----	-----	: 80			
AtCDAR7	-----QVNEDE	CA	MSYSLV	MSLAL	DNKIS	TPYPL	SLA	-----	-----	: 92			
OsCDAR1	QENEDIDRAIAIAE	-----QECSISG	XKMPV	LDG	-----QIARAI	CESLV	MSVMSR	GHG	-----HMYGVNGVANGDIYVPRITFCMDPR	CAGNMME	: 92		
OsCDAR2	FQNEIDRAIAIAE	-----QYKKAIE	KMH	LDG	-----QIARAI	CESLNIV	SPSPR	-----ENGNGANG	CNMYCPLPFPSSSGFR	CAGHSEIG	: 87		
OsCDAR3	HNNEMHATAIAIAA	-----KNYKRNHDK	-----GNNQ	-----QIARAI	CDSLN	NPYCP	-----Y	-----NCPASCP	-----CARSGRY	CAGKHTEIG	: 77		
OsCDAR4	SHHEDIDRAIAIAE	-----QNKKA	VADVED	DIHRL	LDG	-----QIARAI	CESLN	DEPRGN	-----IPVNVDSHST	PTATFPMPFIYFSTGLR	CAGCKPTHG	: 93	
OsCDAR5	SNHEDIDRAIAIAE	-----QNKKA	VADVED	DIHRL	LDG	-----QIARAI	CESLN	ADSPICRN	-----TPVKNVPE	-----PPREL	PLPPIYFSTGLR	CAGCKNTHG	: 88
Pp3c25_330	PSSEEDIRAI	IAIAEE	YQSKRGKQFPVNN	LDG	-----QIARAI	CESLV	GHGRPY	-----AYSSS	-----YAPFPSSSGMNN	CAGGESIG	: 81		
Pp3c16_810	PSLEEDIRAI	IAIAEE	EEYQSKSGKGFAPVNN	LDG	-----QIARAI	CDSV	FGHGRPWV	-----PDYTPS	-----YPPFPARSGLTG	CAGGESIG	: 84		
Pp3c1_850	ASMEATHTAI	IAIAE	-----NRGSKHMS	-----T	-----DIARAI	CDSL	QYLRK	-----EKSPS	-----TRAVFLKPTANN	CAGCKPHG	: 78		
Pp3c13_1250	FGKRGSGYGVYGVGK	CEASVY	RYKTEG	-----YRSG	-----DIARAI	CDS	THIRVYSVNN	-----DAPMALP	PLIPVAPANN	GLVSLG	CAGHRTIG	: 88	

[illegible][illegible]

	DAR-ZF	DUF3633	
s1dAR1	RRPI-G-GYRIID-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRPI-SFPIVEVEIGICQIAHNIWDSFAGSGSTTSASTAS--S	377
s1dAR2	RRPM-G-GREVVCIRHQPLIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRNL-SFPIVEVEIGICQIYWIWESFWM-PGSRMNPSTAS-TA--S	366
s1dAR3	RRPI-GAGNVRMD-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRTSQFVEVEIGICQIAHNIWLETQIASIS--SSNG--	375
s1dAR4	RRPKF-G-GRLGDIRHQPLIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRNTLTFVEVEIGICQIASHWIESEFV-PGSRMNPSTAS-TA--S	364
AtdA1	RRPI-GAGYLLIG-IEPECLIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRNTRFEVEIGICQIAHNIWESSTYAGS--TLVDIAS--	373
AtdA2	RRRL-G-AHRLIV-RQCQILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRNNFEVEVEIGICQIYWIWESLSDPNRLSTSSVA--T	363
AtdA3	GERMGMNKLITDIEVSECVS-GFIVAGITLLVGLPHG--ESTSFIFVHLIT	RRKNLKELEGLCCQGLRWESCTFASDAAAAAAVASSSSSSFS	360
AtdA4	GVR-MLNKILITDIEVSECVS-GFIVAGITLLVGLPHG--ESTSFIFVHLIT	RRNNMVLVEGLCCQGLRWESCTFASDAAAAAAVASSSSSSFS	362
AtdA5	ERST-GSKNKLITDIEVSECVS-GFIVAGITLLVGLPHG--ESTSFIFVHLIT	RRNNMVLVEGLCCQGLRWESCTFASDAAAAAAVASSSSSSFS	362
AtdA6	RRVR-GPNGLIVG-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRNNNILEGLCCQIICHWIDWCTATADADASSASS--SS	374
AtdA7	KFTM-GPNGLIVS-GPEQIVYVGCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRKNLKELEGLCCQIICHWIDWCTYSS--SSAASSASS--SS	375
AtdA1	RSKH-GTKWAGN-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRTSQFVEVEIGICQIAHNIWDLDAAGS--TNSNA--	375
CsDA1	RRFM-AGNVRMD-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRTSFPIVEVEIGICQIAHNIWESSTIAGS--GNSGAS--T	372
CsDA2	RRPI-G-ANRLID-RQCQILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRNTRKAEPIEGICQIYWIWESLSTSRYQGASTSYAS--S	368
CsDA3	RRVI-GPGNIGID-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRQIDPKFVEVEIGICQIAHNIWESLTGSS-SIASIASIASS--S	383
CsDA4	RRIL-GPGNIGID-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRNSFPIVEVEIGICQIASHWIESEFTGS-SIDIATSYAS--S	378
ppc3c1_330	GGK-GRNVRMD-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRPRFPIVEVEIGICQIASHWIESEFTGS-SIDIATSYAS--S	378
ppc3c1_8	GGK-GRNVRMD-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRPRFPIVEVEIGICQIASHWIESEFTGS-SIDIATSYAS--S	378
ppc3c1_850	GCKS-SFPRLSFNW-IEPEYILIRRCISVITLLVGLPRLRTGIIAHENMHAWLAHLK	RRPRFPIVEVEIGICQIASHWIESEFTGS-SIDIATSYAS--S	378
ppc3c1_120	NGG--ERPLIRRCISVITLLVGLPHG--ESTSFIFVHLIT	RRPNENFEVEIGICQIAHNIWAKENLIT	357

DUF3633

S1DAR1	SSSSSPSTSRKKGKRSDEPKKIGLDFKKQIESDSAPYGLGFRFGNKAVL--KYGLRKTIDHRLTGTGYPC----	: 446
S1DAR2	SSS---MSSTRKGVKSGVNNKIGEFEMQIAHDSIPAYGGGFRSANPAVN--KYGLRCTIDHRLTGSFFEL----	: 432
S1DAR3	ATSSSGMSSKQGISPFERKIGLFEKHQIESDTSPIYGGNFRAGNQAVL--KYGLRSTIDHRLTMTGTFPY----	: 444
S1DAR4	SSST---WSSRKGGKSQAASKLGEFEMQIAHDSIPAYGGGFRAAYPAAVN--KYGLRSTIDHRLTGNFPEL----	: 431
AtDAR1	SSSAVVSASRKGGERSDEPKKIGEFKKQIESDSSAPYGLGFRQGNQAVL--KHGLRRTIDHRLTGTTFP----	: 441
AtDAR2	SSSS---SFSNRKGGKSNVKKIGLDFEKQIAHDSIPAYGGGFRANAAC--KYGLRRTIDHRLTGTTFEL----	: 430
AtDAR3	TAPPAAIITKKSDDWIEFKKKIGEFCKMKQIKEDSPVYGLGFKQVYEMVSNNNIKDITKDVVSNANTPDSTV	: 435
AtDAR4	RTPLSTTITKKSDVPEDEKFRIVNCKQIKETIDSPFFGIGFRKVNKMNASNHSLKDTKEILISIKTEQYSKL	: 456
AtDAR5	RTPPAASASRKGE-WSDDFKKIVFECKNQIETDSPVYGLGFRFTVNMNT--NSSLQETIKELRLRR-----	: 438
AtDAR6	RTPPAASASRKGE-WSDDFKKIVFECKNQIETDSPVYGLGFRFTVNMNT--NSSLQETIKELRLQR-----	: 438
AtDAR7	RTPAANAASRKGA-QSDYFKPKIVFECKQIETDSPVYGVGRKVNQMVS--DSHLHKISQIHWTKPDSNL-----	: 444
AtDAR1	ASSSSSSQGLRKGPRESQYRKIGLDFKKQIESDSPVYGLGFRAGRLAVH--KYGLRKTIEHRLTGMTGRFFV----	: 444
OsDAR1	SSSSSSASTSRKGGERSQERKIGLDFKKQIESDSAPYGLGFRAGNRAVL--CYGLRRTIEHRLTGTTFEF----	: 444
OsDAR2	SSSSCRPPSSKGGVISHTPKKIGLDFEHLQIANDSSAPYGLGFRAAYPAAVN--KYGLRQSNHRLTGTGFFV----	: 437
OsDAR3	SSSSSAPASRKGVIGLDFPKKIGEFKKQIETDSDVYGLGFRDGIRAVE--RYGLRRTIDHRYLTGVFFPC-----	: 452
OsDAR4	SSSSSTPTSRKGAKLEEPKKIGAFIKQIETDSSAPYGLGFRANRAVE--SVGLRSTIDHRYKMTGTFPY----	: 447
Pp3c25_330	---GARSSNS--AAISARIGRYLYLQISSDSFPVYGLGFRHGMAAVQ--QGLSERVLDHRLTGNFPEL----	: 433
Pp3c16_818	---GRSSSSSSSSQAIATRIAPAEYLYLQIASSDSFPVYGLGFRGYIAVQ--PFGSERVEHRLTANFPEL----	: 440
Pp3c1_8550	---KKDSTGAG-INSFAPGERIGKYLYLQISSDSFPVYGLGFRQALASVN--RYGLVRVPEHRTMTANFQV-----	: 422
Pp3c13_120	---RTTSGDGS---SVSKRIGFELYQIETISSQYGLGFRASKAVA--XGRLPTTHHRLTGTGEILR-----	: 419

Fig. 2. (continued).

3.3. Sequence analysis

Genomic DNA sequence analysis showed that all genes had five exons and four introns except *SIDAR1*, 3, 6 and 9 (Supplementary Fig. 1). Intron 2 was relatively short compared to the other introns, and introns 1, 3 and 4 were highly variable in length (Supplementary Fig. 1). The first, second, third, fourth and fifth coding regions were 135–138 bp, 97–103 bp, 20–44 bp (shorter than the other coding regions), 90 bp (most conserved among all LIM genes) and 207–333 bp (longer than the other four) in length, respectively (Supplementary Fig. 1). The mRNA sequences of *SIDAR1*, 2, 3, and 4 were relatively long, with a higher number of introns and coding sequences than the 11 other genes (Supplementary Fig. 1). In the CRP-like protein family, the LIM1 domain was located in exon 1 and 2 and the LIM2 domain is located in exon 4 and 5 (Supplementary Fig. 1). In the DAR family, the LIM domain was located in exon 4 and 5, the UIM domain was located in exon 1 and 2 and the DUF3633 domain was located between exon 7 and 11 (Supplementary Fig. 1). However, in *SIDAR4*, the LIM domain was located in exon 5 and 6 and the DUF3633 domain was located in exons 8 to 12 (Supplementary Fig. 1). Based on homology analysis, tomato LIM proteins shared high sequence similarity (60–89%) with LIM proteins from Arabidopsis and other plant species (Supplementary Table 3). Fifteen LIM proteins exhibited a wide variety of sequence identity between each other ranging from 20 to 88% (Supplementary Table 4). However, sequence identity within subfamily proteins was comparatively higher that ranged from 53 to 74% among four DAR proteins and 45–88% among 11 CRP-like proteins of tomato (Supplementary Table 4). The sequence identity of the proteins from DAR subfamily never exceeded 44% in

comparison with any of the proteins from CRP-like subfamily (Supplementary Table 4). The highest sequence similarity of 88% was noted between *SlβLIM1a* and *SlβLIM1b* (Supplementary Table 4).

Multiple protein sequence alignment of tomato LIMs with other plant LIM proteins in the CRP family revealed that the LIM1 and LIM2 domains are separated by an inter-LIM spacer (Fig. 2a). Distinct amino acids, including most conserved cysteine-histidine residues, were conserved in the LIM domain and in other regions of the sequences (Fig. 2a). The regions immediately flanking the LIM1 and LIM2 domains show sequence conservation, i.e. the N-terminus as well as about 14 residues C-terminal of LIM2 are well conserved. Further, the Inter LIM region shows conservation of about 15 residues C-terminal to LIM1, and 12 residues N-terminal to LIM2 (Fig. 2a). Protein sequence alignment of members of the DAR family revealed one LIM domain with one or two UIM motifs and one DUF3633 domain (Fig. 2b). Little similarity was found between the *SIDAR* group and the *SlLIM* and *Wlim* groups based on protein sequence alignment, which differentiates the *SIDAR* group from these LIM domain-containing groups (Fig. 2a and b). A highly conserved putative zinc finger motif (around 10 Cys-His residues) was found in between the LIM and DUF3633 domains in the plant-specific DAR family (Fig. 2b). Previously, Park et al. and Zhao et al. identified conserved sequence features at the C-terminus of DAR LIM proteins (Park et al., 2014; Zhao et al., 2014); Park et al. (2014) also noted conserved cysteine and histidine residues. Peng et al. also indicated a putative zinc binding domain in the C-terminus without giving any sequence, however (Peng et al., 2013). Our multiple sequence alignment with the tomato DAR LIM protein sequences confirmed this highly conserved C-terminus. Part of this

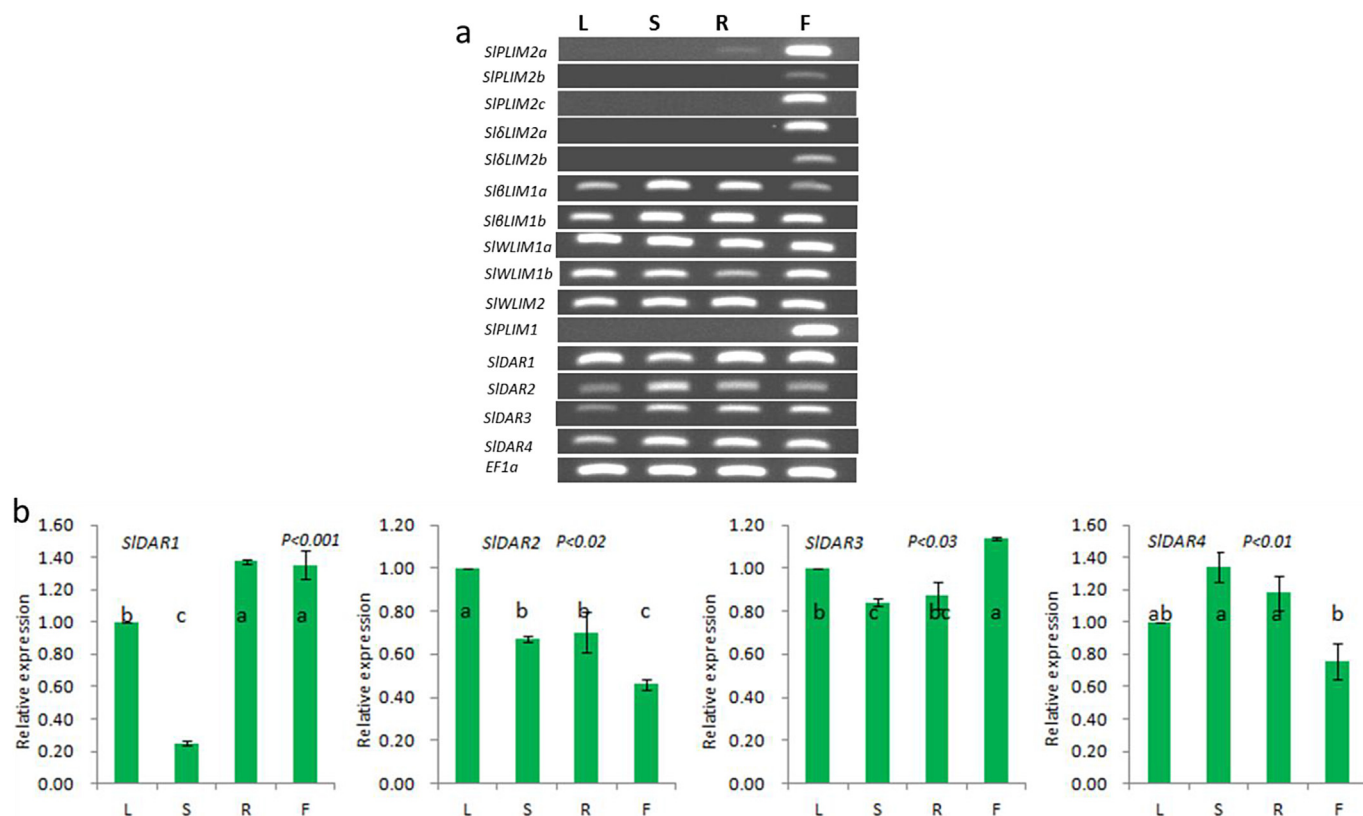


Fig. 3. RT-PCR analysis of LIM gene expression in different organs of *Solanum lycopersicum* a) L-leaf, S-stem, R-root and F-flower tissue. Transcript accumulation was recorded after 30 cycles of PCR, except for *EF1a* (27 cycles) and *SIDAR2* and *SIDAR4* (32 cycles). b) qPCR expression analysis of DAR family LIM genes in L-leaf, S-stem, R-root and F-flower tissue. The error bars represent the standard error of the means of three independent replicates. P-values indicate statistical significance. Different letters indicate mean values at different sampling points after Tukey's pairwise comparison test.

C-terminus (marked in green, Fig. 2b) is now identified in PFAM (<http://pfam.xfam.org/family/PF12315>) as domain of unknown function (DUF3633), which is found not only in plants, but also in lower eukaryotes and apparently in bacteria. However, the region between DUF3633 and the LIM domain is also highly conserved, containing 10 conserved cysteine and histidine residues (marked in yellow, Fig. 2b), suggesting that this might be a domain with possibly at least two zinc fingers. Blast searches and domain analysis showed that this is not a known domain, but is conserved in plants. We suggest to name this domain “DAR-ZF”. We also used MEME to try to identify new motifs in the protein sequences, though none were found that were not obvious in the MSA (Supplementary Fig. 2 and Supplementary Table 5a,b). First two blocks of Fig 2 should be marked with a (Fig 2a) and the following 5 blocks should be marked as b (Fig 2b).

3.4. Chromosomal localization and duplication analysis of tomato LIM genes

The 15 identified tomato LIM genes were located on different chromosomes. All chromosomes contained at least one LIM, except for chromosome 7 and 9. The distribution of genes on the chromosomes appeared to be uneven, with most genes residing on the ends of chromosomes (Supplementary Fig. 3). Chromosome 5

contained three LIM genes, chromosome 3 contained two and the other chromosomes contained a single gene (Supplementary Fig. 3). *SIDAR2* and *SIDAR4* were predicted to be segmentally duplicated because the identity and percent query coverage of these two genes were more than 80% (Kong et al., 2013). These two genes were located on chromosome 11 and 12, respectively (Supplementary Fig. 3). No tandemly duplicated genes were found among the 15 LIM genes, as no genes located within a range of 100 kb on the same chromosome were detected (Supplementary Fig. 3).

3.5. Organ-specific expression of tomato LIM genes

Expression analysis revealed that *SIPLIM2a*, *2b*, *2c*, *SlδLIM2a*, *SlδLIM2b* and *SIPLIM1* were expressed only in flowers, whereas the other five genes of the CRP-like subfamily were expressed at variable levels in all organs. *SlβLIM1a* and *1b* were more highly expressed in stems and roots than in leaf tissue (Fig. 3a, b).

Among DAR subfamily members, all four genes were expressed in all organs, but their expression levels varied in different organs (Fig. 3a, b). *SIDAR1* and *SIDAR3* were more highly expressed in flowers than in leaves, but *SIDAR2* and *SIDAR4* were more highly expressed in leaves than in flowers (Fig. 3b). We therefore selected five genes from the CRP-like subfamily (*SlβLIM1a* and *1b*, *SIWLIM1a*

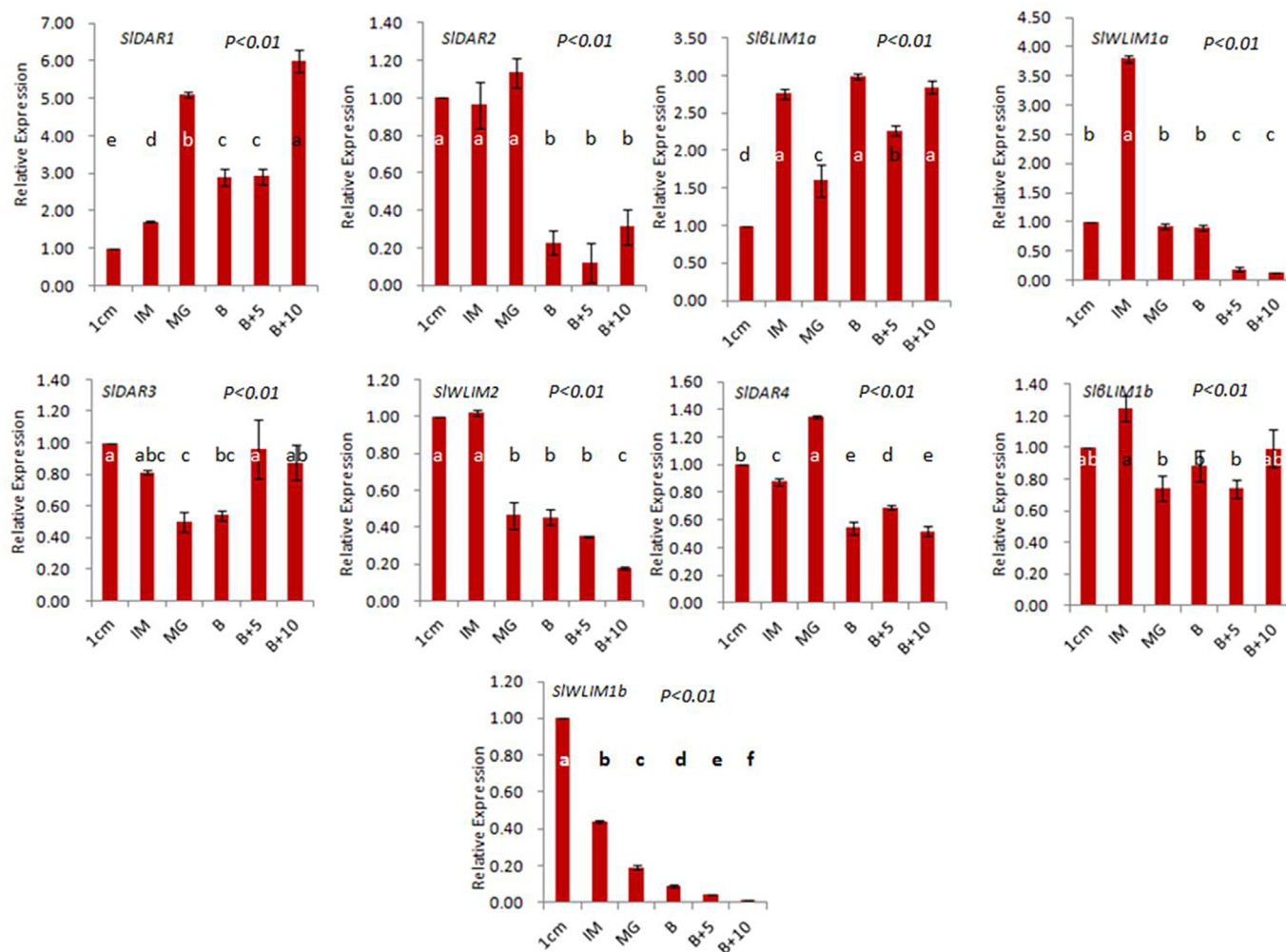


Fig. 4. qPCR expression analysis of LIM genes in fruit at six developmental stages (1 cm–1 centimeter fruit, IM–immature fruit, MG–mature green, B–breaker, B+5–5 days after breaker and B+10–10 days after breaker). The error bars represent the standard error of the means of three independent replicates. P-values indicate statistical significance. Different letters indicate mean values at different sampling points after Tukey's pairwise comparison test.

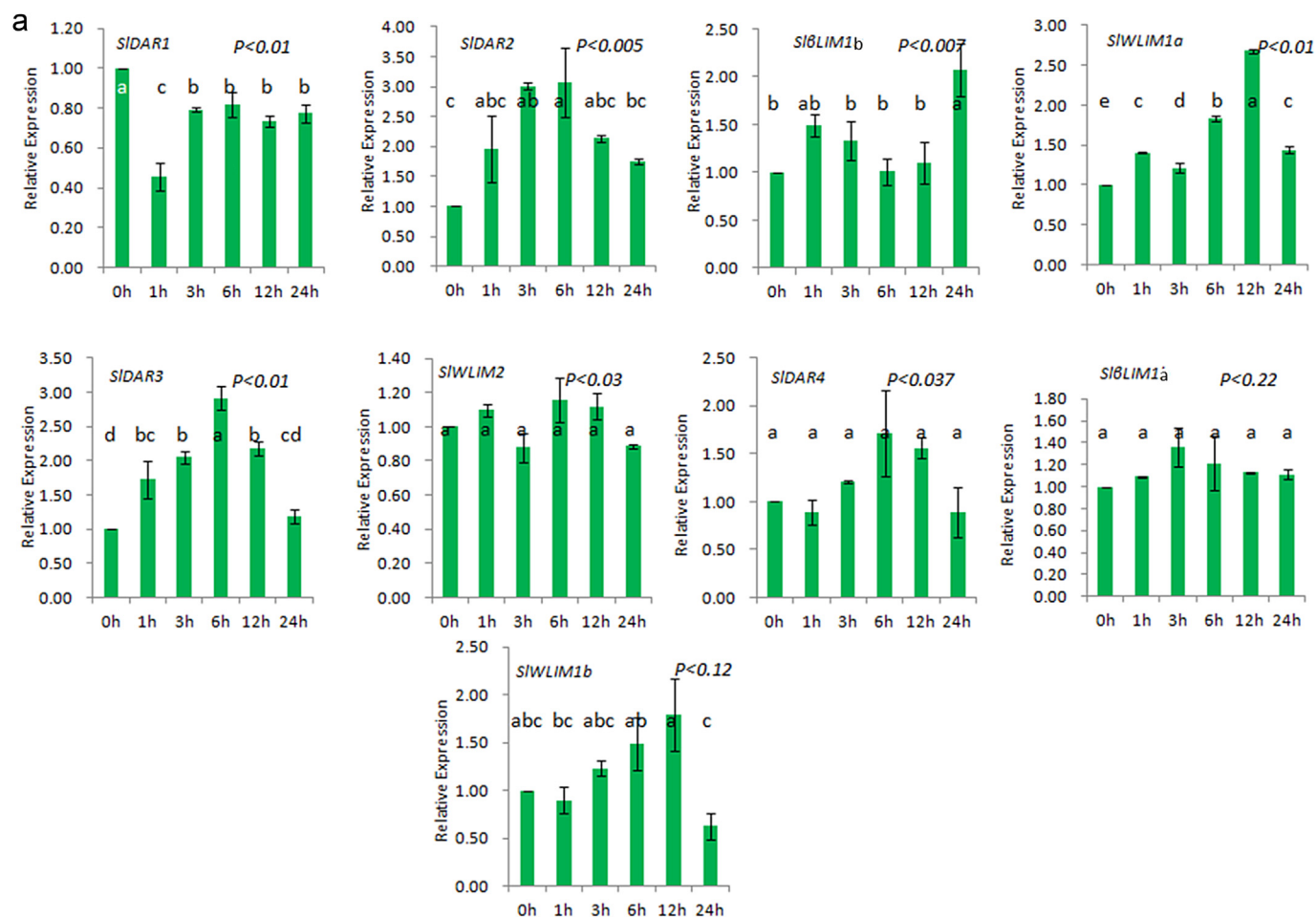


Fig. 5. q-PCR expression analysis of nine tomato *LIM* genes in response to a) ABA b) drought c) cold d) NaCl and e) heat treatment at 0, 1, 3, 6, 12, 24 h of treatment. The vertical bars represent the standard error of the means of three independent replicates. P-values indicate statistical significance. Different letters indicate mean values at different sampling points after Tukey's pairwise comparison test.

and *1b* and *SIWLIM2*) and all four genes from the DAR subfamily for expression analysis in fruits at different stages and in plants under different abiotic stress treatments. *SIDAR1* and *SIβLIM1b* were highly expressed at the later stage of fruit development, whereas *SIWLIM2*, *SIDAR2* and *SIDAR4* were highly expressed at the early stage of fruit development ($p < 0.01$, Fig. 4). *SIWLIM1a* was highly expressed in immature fruit (IM) compared to young fruits (fruits 1 cm in diameter) and its expression level decreased at later stages of fruit development ($p < 0.01$, Fig. 4). *SIWLIM1b* was highly expressed in young fruits, and its expression gradually declined, with almost no expression detected at 10 days after the breaker stage of fruit development ($p < 0.01$, Fig. 4). The expression of *SIDAR3* gradually decreased up to the mature green (MG) stage and then increased after the breaker stage of fruit development ($p < 0.01$ Fig. 4). The expression of *SIβLIM1a* was slightly downregulated after the immature stage (IM) to the breaker stage of fruit development ($p < 0.01$, Fig. 4).

3.6. Expression analysis under abiotic stresses conditions

Seven of the nine genes were deregulated by ABA treatment (Fig. 5a). *SIDAR2*, 3 and *SIWLIM1a* were significantly upregulated by this treatment, but *SIDAR1* was downregulated compared to the control (Fig. 5a). The expression of *SIβLIM1b* increased slightly up

to 3 h after treatment and then declined at 6 h and 12 h, followed by an increase at 24 h after treatment (Fig. 5a). *SIDAR4* and *SIWLIM1b* were upregulated up to 12 h after treatment and downregulated at 24 h after treatment compared to the control (Fig. 5a). Seven of the nine genes were differentially expressed under drought stress treatment (Fig. 5b). *SIDAR3* and *SIDAR4* were upregulated but *SIDAR2* and *SIWLIM1b* were downregulated compared to the control (Fig. 5b). *SIβLIM1b* was gradually upregulated up to 6 h from the start of treatment and downregulated thereafter (Fig. 5b). *SIWLIM1a* and *SIWLIM2* were highly expressed only at 12 h of treatment compared to the control (Fig. 5b). Five of nine genes were differentially regulated by cold treatment (Fig. 5c). *SIDAR3*, 4 and *SIWLIM2* were upregulated compared to the control (Fig. 5c). *SIβLIM1a* and *SIβLIM1b* were upregulated until 12 h after treatment and then downregulated at 24 h compared to the control (Fig. 5c). Eight genes out of nine were deregulated by NaCl treatment, including six genes (*SIDAR1*, *SIDAR2*, *SIDAR4*, *SIβLIM1a*, *SIWLIM1b* and *SIWLIM2*) that were downregulated compared to the control (Fig. 5d). *SIβLIM1b* was downregulated until 12 h after treatment and was upregulated at 24 h after treatment (Fig. 5d). *SIWLIM1a* was upregulated at 3 h after NaCl treatment compared to the control (Fig. 5d). Seven genes were significantly differentially regulated by heat-stress treatment (Fig. 5e). *SIDAR1*, *SIWLIM2* and *SIβLIM1a* were downregulated

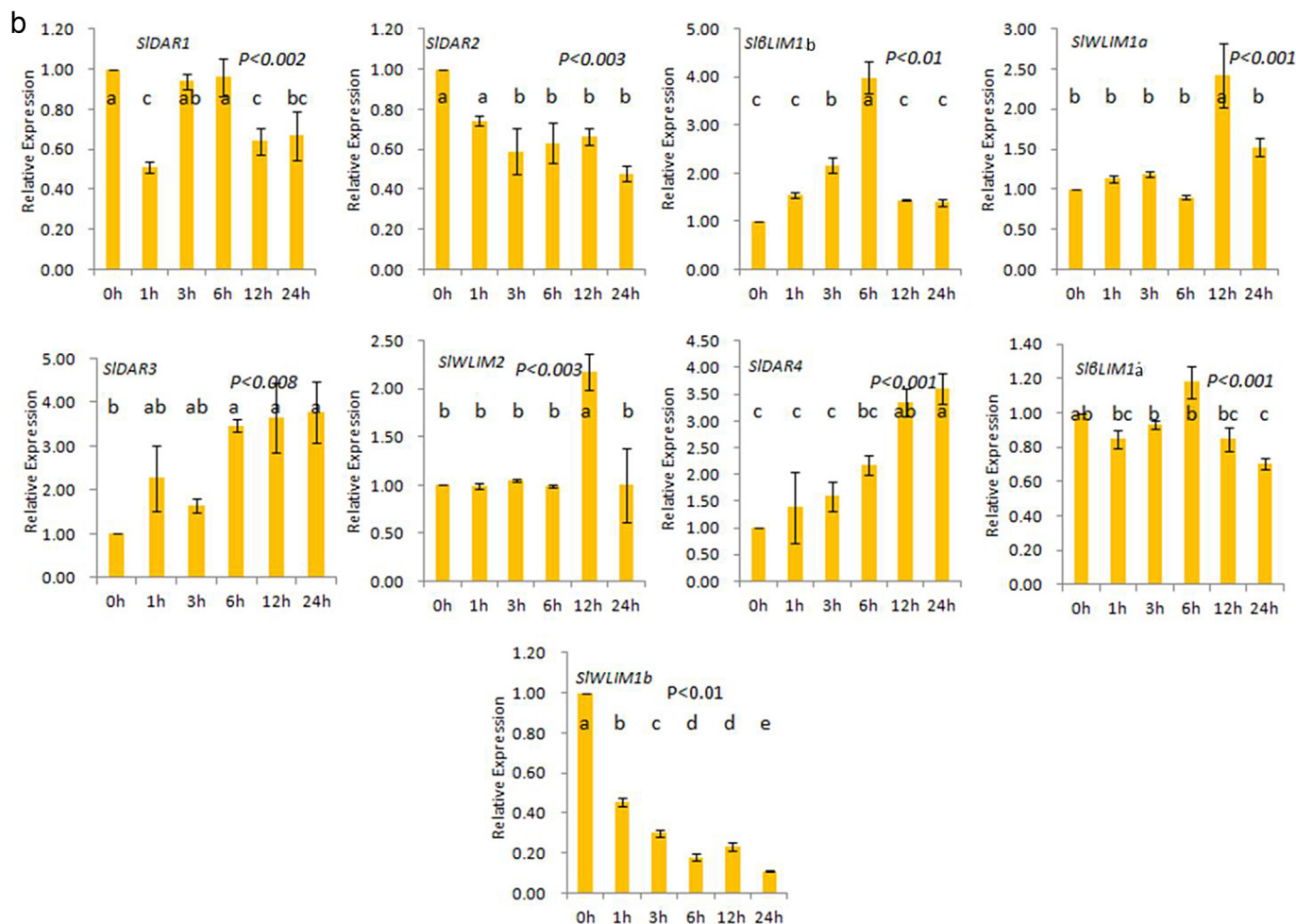


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compared to the control (Fig. 5e). *SIβLIM1b* was highly expressed at 1 h after treatment, followed by a decline compared to the control (Fig. 5e). *SIWLIM1b* was downregulated at 12 and 24 h after treatment compared to the control (Fig. 5e). Finally, *SIWLIM1a* and *SIDAR3* were upregulated at 3 h after treatment compared to the control (Fig. 5e).

4. Discussion

In the present study, we identified 15 tomato LIM proteins containing structurally similar domains to those of other LIM protein family members (Arnaud et al., 2007). The tomato LIM proteins share more close phylogenetic relationships with tobacco (CRP-Like family) and Chinese cabbage (DAR family) family members, with high bootstrap support, implying their close evolutionary correlation. Some of the branch bootstrap values were found to be insignificant in the phylogenetic tree, probably due to divergent sequences. In vertebrates, the CRP-like LIM proteins are involved in cytoskeleton remodeling, transcriptional regulation and cell differentiation, and they appear to have evolutionarily conserved functions in plants (Weiskirchen and Günther, 2003). Therefore, CRP-like LIM proteins may play the same structural roles in all cell types. Phylogenetic analysis clearly confirmed the existence of the PLIM1, PLIM2, WLIM1, WLIM2 and βLIM1 groups in tomato, which were also reported in other crop plants (Eliasson

et al., 2000; Arnaud et al., 2007). The recently identified βLIM1 group forms a distinct clade in the phylogenetic tree (Arnaud et al., 2007; Eliasson et al., 2000; Park et al., 2014). The distinct clade of the three moss LIM proteins within the CRP-like LIM family in the phylogenetic tree implies that the split into different families (PLIM2, WLIM2, WLIM1, PLIM1 and βLIM1) occurred later in evolution after the split of mosses from other land plants. The multiple sequence alignment revealed a novel zinc-finger type domain, DAR-ZF, between the LIM and the DUF3633 domain that had so far not been commented upon (Park et al., 2014; Zhao et al., 2014). The origin of DAR-ZF is presently not clear, whether it is derived from LIM fingers, or other zinc fingers, though it seems well conserved in plants. Nothing is known about the function of the DUF3633 domain, but it is ancient, as it is found in bacteria, which suggests that it might have merged at some point in evolution with a LIM domain protein.

Spatial or temporal expression profiling of a gene during growth and development and under stress provides clues about its possible function, which forms the basis for future functional studies of the gene. The differential expression patterns of the 15 LIM genes may be helpful for exploring their functions in tomato. The expression patterns of the 15 LIM genes are also strongly associated with their phylogenetic classifications (Figs. 1 and 3a and b). Six genes among the 15 examined are expressed only in flowers. These genes were clustered with the PLIM1 and PLIM2 gene subfamilies of other plant

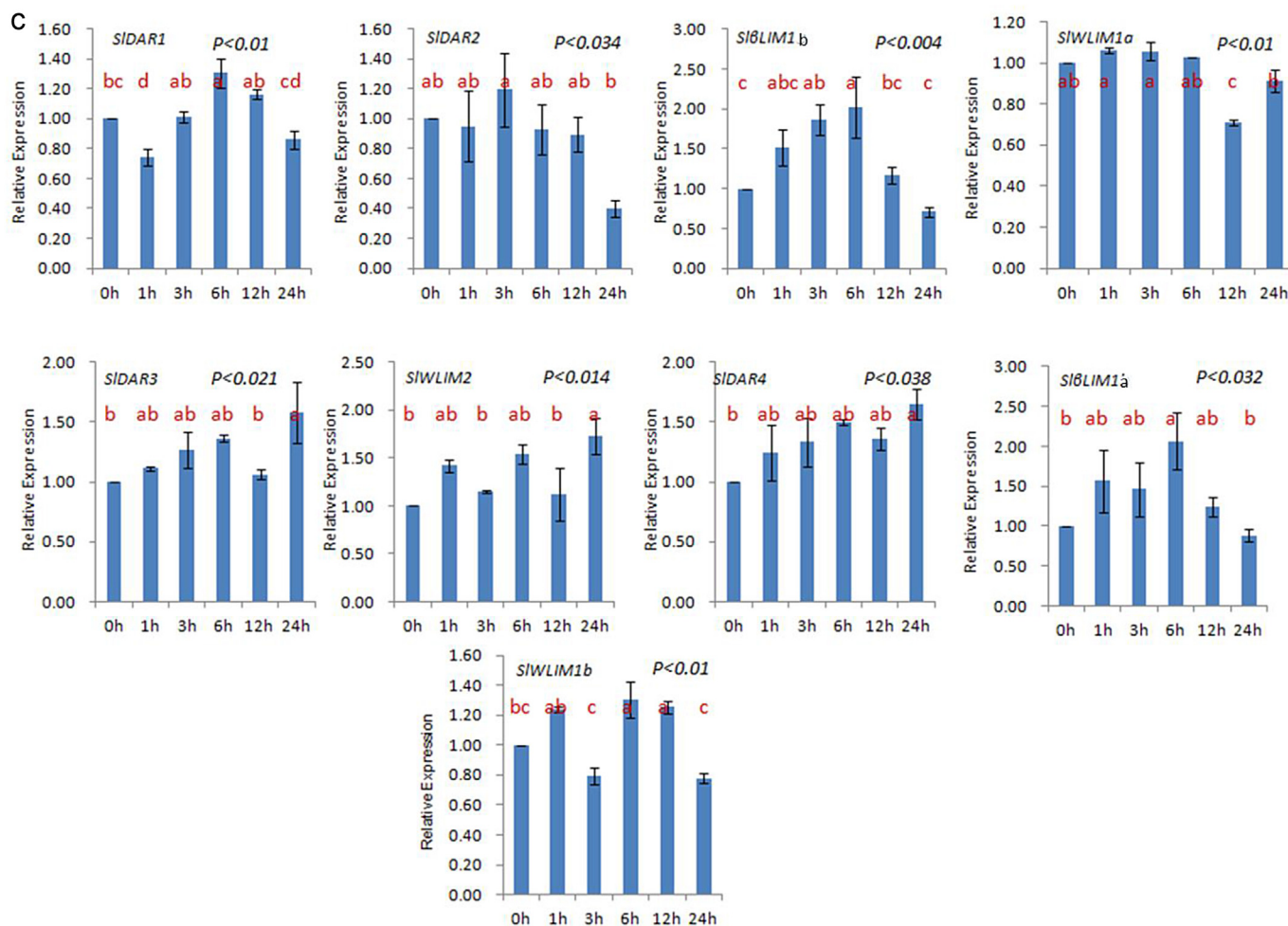


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species in the phylogenetic tree, suggesting their possible roles in flower and pollen development (Figs. 1 and 3a and b). In some plant species, the *PLIM1* and *PLIM2* family genes are pollen specific and likely function in pollen growth and development. For example, in sunflower, *HaPLIM1a* is highly expressed in maturing pollen, suggesting its possible role in pollen tube growth and pollen germination (Arnaud et al., 2007). Similarly, in *Lilium longiflorum*, *LiLIM* may play a role in pollen tube growth (Wang et al., 2008) and in *Nicotiana tabacum*, *NtPLIM1* may function in pollen germination and/or pollen tube growth (Sweetman et al., 2000). The nine other genes were ubiquitously expressed in all organs (Fig. 3a). Moreover, they formed a cluster with the *WLIM* group in the phylogenetic tree, suggesting that these genes are members of the *WLIM* sub-family, which is widely expressed in all sporophytic plant tissues (Eliasson et al., 2000; Wang et al., 2008). These genes are thought to be involved in growth- and development-related processes in different organs. *SIβLIM1a* and *SIβLIM1b* were more highly expressed in stem and root tissues than in other organs (Fig. 3a). Stem tissues strengthen the plant and are involved in long-distance transport of water and nutrients, thus allowing plants to grow and withstand stress. Stem tissues also provide structural support through the deposition of lignin. Therefore, the higher expression of *SIβLIM1a* and *SIβLIM1b* in stems suggests that they may function in regulating lignin biosynthesis via the phenylpropanoid pathway, although this requires further functional analysis. The regulation of lignin biosynthesis by LIM has been reported in *Nicotiana*,

Eucalyptus and *Gossypium* (Kawaoka et al. 2000, 2006; Kawaoka and Ebinuma, 2001; Han et al. 2013).

The role of LIM genes in fruit development has not previously been explored in any vegetable species except *Glycine max* (Zhao et al., 2014). Tomato fruit development involves two steps: fruit enlargement and fruit ripening (Giovannoni, 2007). During fruit enlargement, both cell division and cell elongation occur until the mature green stage, when the fruits obtain their maximum size. Ripening then begins at the breaker stage through the formation of carotenoids via the degradation of chlorophyll. *SIDAR2*, *SLDAR4* and *SIWLIM2* are expressed at relatively high levels before the breaker stage, suggesting that they might be involved in cell division and elongation and might therefore function in increasing the size of fruit (Fig. 4). Ripening of tomato fruits depends on ethylene levels, which increase from the breaker stage to the orange stages (Giovannoni, 2004). Our results suggest that *SIβLIM1b* might be involved in fruit ripening. However, this assumption requires further functional validation. Zhao et al. (2014) also speculated that LIM genes function in fruit development.

As actin is a vital signaling molecule, the actin cytoskeleton can recognize various abiotic and biotic stimuli (Smertenko and Franklin-Tong, 2011; Day et al., 2011). Many plant LIM proteins (e.g., lily *LILIM1*, tobacco *WLIM1* and Arabidopsis LIM proteins) regulate filamentous actin bundle assembly, protect their depolymerization and control actin dynamics (Papuga et al., 2010; Wang et al., 2008). Thus, the actin binding and bundling capacity

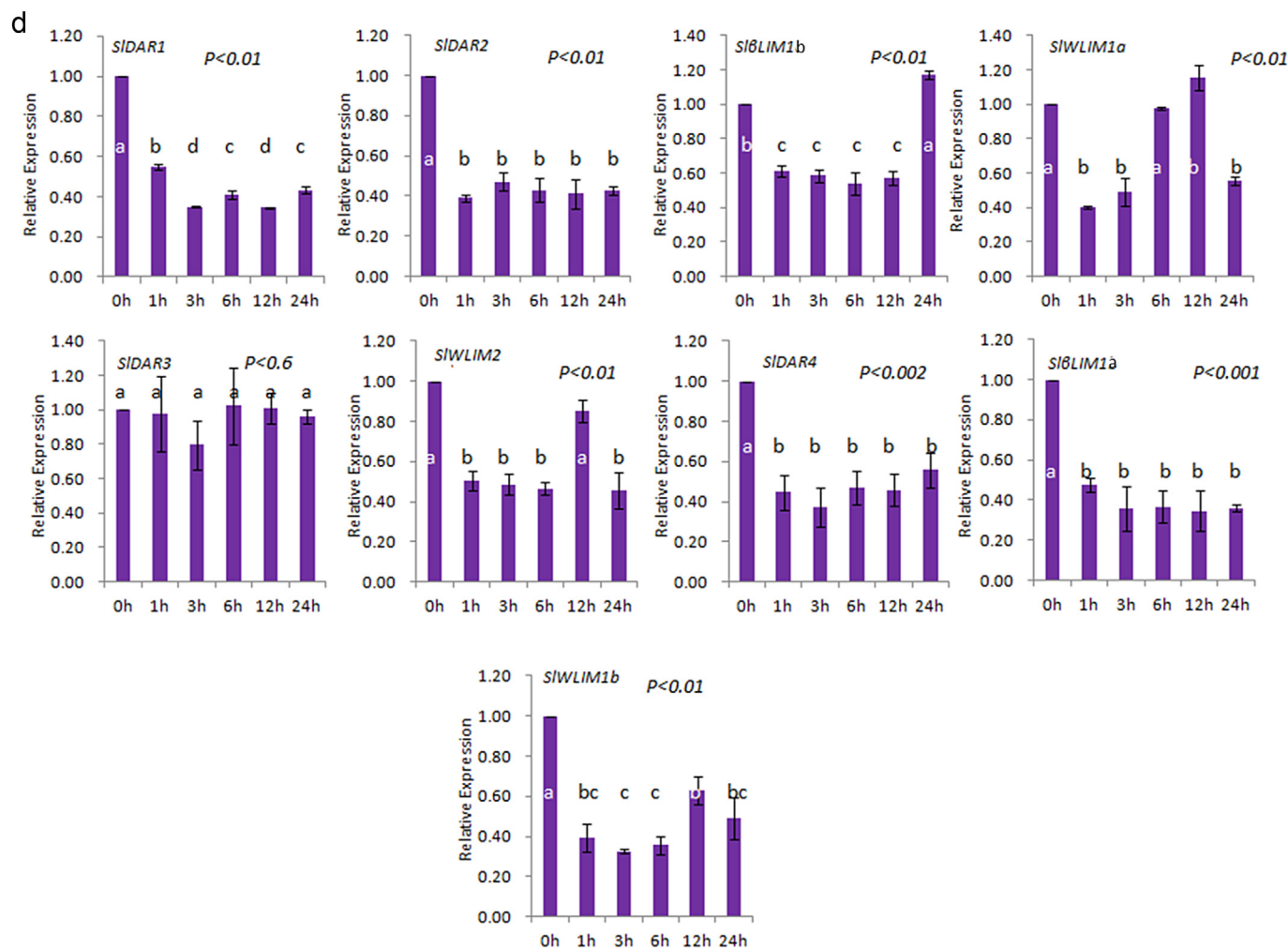


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of the CRP-like LIM protein may act as a mechanism for abiotic stress responses. Arnaud et al. (2012) suggested that PtLIMs might function in the response against mechanical stress. To date, the role of *LIM* genes in abiotic stress responses has not been explored in tomato or in any other solanaceous crop. Differential expression patterns of tomato *LIM* genes were detected (both upregulation and downregulation) under ABA, drought, cold, NaCl and heat stresses. These changes in the expression patterns of these genes in response to abiotic stress indicate their likely association with abiotic stress tolerance.

The DAL and DAR family LIM proteins are plant specific and are involved in regulating actin dynamics and organ size, as well as defense responses (Zhao et al., 2014). In addition to LIM domains, these proteins have other domains, such as the UIM and DUF3633 domains and a conserved C-terminus, which structurally and functionally differentiate this family from the CRP-like family. The UIM domain interacts with ubiquitin and functions in the ubiquitination process (Xia et al., 2013). Ubiquitination regulates various cellular processes, including stress response and DNA repair (Huang and D'Andrea, 2006). In the current study, all four *SIDAR* genes exhibited differential expression patterns under five different stress treatments. Moreover, abiotic stress-responsive promoters/cis-acting elements were found in all 15 tomato *LIM* genes, which may help regulate their expression under different abiotic stress conditions (Supplementary Table 6).

5. Summary

In this study, we identified 15 *LIM* genes in tomato. Tomato LIM proteins form two distinct functional groups, the CRP-like proteins and other plant-specific DAR proteins, which are similar to LIM protein families in Arabidopsis and other plants. We analyzed the expression of these genes in different organs, including different stages of fruit development, revealing that *SIDAR2*, *SIDAR4*, *SIβLIM1b* and *SIWLIM2* are promising candidates for further functional analysis. The expression pattern of *SIβLIM1b* suggests that it functions in fruit ripening. *SIDAR2*, *SIDAR4* and *SIWLIM2* were expressed at the early stage of fruit development, suggesting their involvement in cytoskeleton development in fruit. *SIDAR1*, 2, 3, 4, *SIWLIM1a* and *b*, *SIβLIM1a* and *b* and *SIWLIM2* were induced under various abiotic stresses, indicating their possible roles in stress tolerance. This study provides new insights for functional characterization of the important but little explored *LIM* gene family, which may be useful for developing abiotic stress-tolerant tomato cultivars.

Author contributions

Mi-Young Chung, Ill-Sup Nou, Chang Kil Kim, Ki-Byung Lim, Min-Bae Kim and Do-Jin Lee and Jong-In Park designed and supervised the work. Mi-Young Chung, Jong-In Park and Nasar Uddin

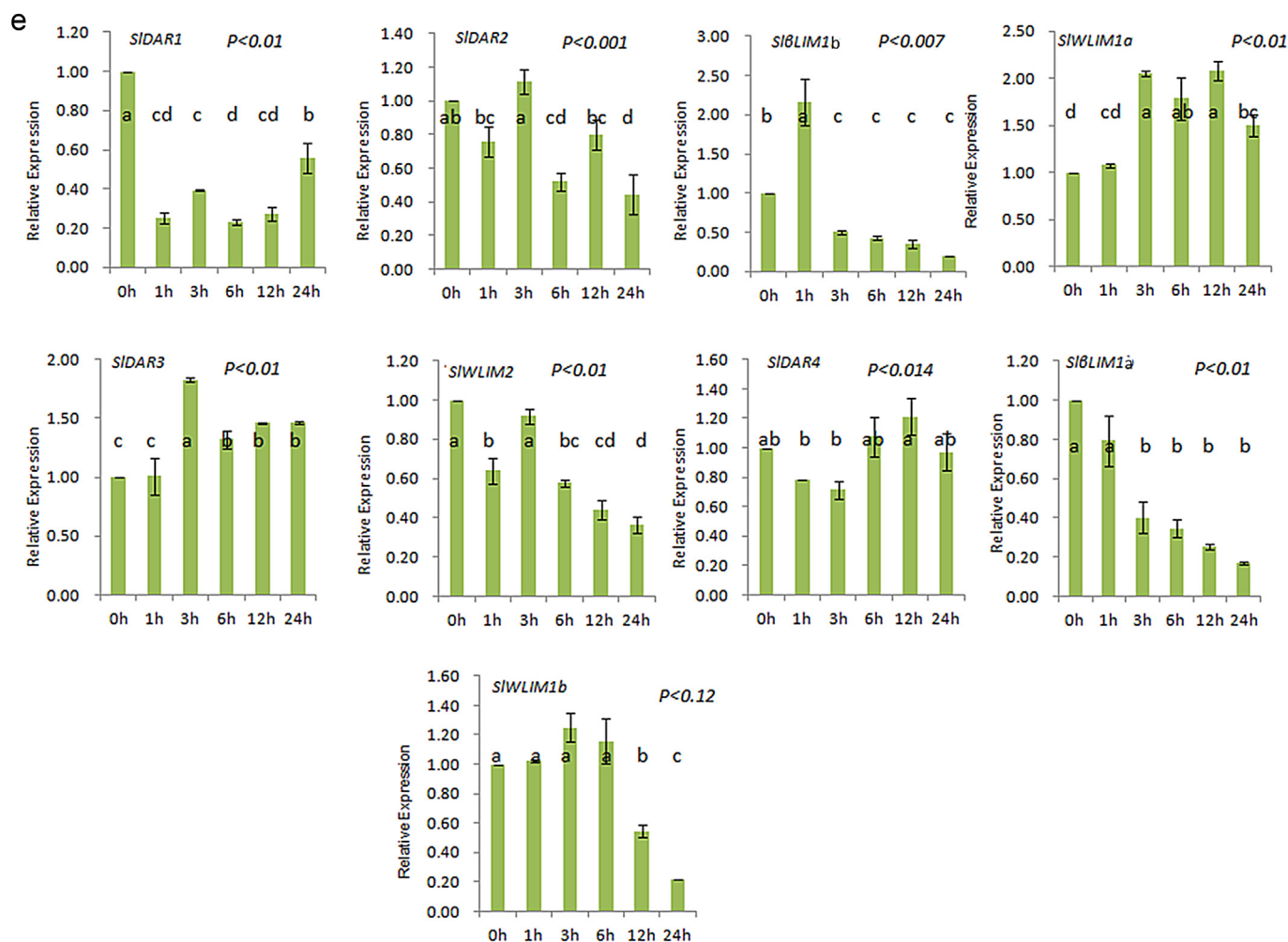


Fig. 5. (continued).

Ahmed monitored the experimental work. Khadiza Khatun conducted in silico analysis, grew plants, isolated RNA and carried out expression analysis as per the guidance of Mi-Young Chung and Nasar Uddin Ahmed. Arif Hasan Khan Robin analyzed the data and extensively revised the initial manuscript prepared by Khadiza Khatun. All authors contributed to the manuscript.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.plaphy.2016.07.006>.

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